

Weizhe Liu

Curriculum Vitae

Department of Astronomy / Steward Observatory
University of Arizona
☎ (240) 413 7164
✉ oscarlwz@gmail.com
ORCID: <https://orcid.org/0000-0003-3762-7344>
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Research Interests

- High Redshift Quasar and Galaxy • AGN and Stellar Feedback.
- Dwarf AGN and Galaxy • Cosmic Baryon Cycle.

Employments

Fall 2022 - present **JASPER Postdoctoral Scholar**, *Steward Observatory*, Supervisor: Prof. Xiaohui Fan.

Education

- 2016–2022 **PhD in Astronomy**, *University of Maryland*, Advisor: Prof. Sylvain Veilleux, Thesis: ACTIVE GALACTIC NUCLEUS FEEDBACK IN GIANTS AND DWARFS.
- 2013–2016 **M.S. in Astronomy**, *Shanghai Astronomical Observatory, University of Chinese Academy of Sciences*, Advisor: Prof. Lei Hao.
- 2009–2013 **B.S. in Astronomy**, *Nanjing University*.

Selected Grants and Awards

- 2026 **NASA/NEExSci grant**, *Keck (NASA) 2026A*, ~ \$11,250, PI.
- 2024–2026 **NASA/STScI grant**, *JWST Cycle 2 Survey 3428*, ~ \$120,000, PI.
- 2025–2027 **NASA/STScI grant**, *HST Cycle 32 GO 17758*, ~ \$64,000, PI.
- 2019–2021 **NASA/STScI grant**, *HST Cycle 27 GO 15915*, ~ \$81,000, PI.
- 2021 **Andrew S. Wilson Prize for Excellence in Research**, awarded to only one graduate student nearing completion of the Ph.D. dissertation within the department.
- 2018 **John Chi-Lin Wang Award for Academic Excellence**, awarded to the 2nd year graduate student with the best overall performance within the department.
- 2012 **Annual Scholarship of National Astronomical Observatory of China**, awarded to the top 5 students each year.
- 2012 **Internship Scholarship of Chinese Academy of Science**, awarded to the best summer intern each year.
- 2012 **Second Prize of Renmin Scholarship**, awarded to the top 2 students each year.
- 2011 **National Scholarship of People's Republic of China**, awarded to only 1 of the top 2 students in the entire astronomy department this year.
- 2010 **Third Prize of Renmin Scholarship**, awarded to the top 6 students each year.

Publications

Citations: first and/or corresponding author (140); total (1209); h-index: 20

- First/Corresp. Author:** [11] **W. Liu**, X. Fan, et al. 2026c, *A Billion Solar Mass Black Hole on the Local $M_{BH-\sigma}$ Relation Quenching Its Host Galaxy at $z \sim 6.2$* , to be submitted by May 2026.
- [10] **W. Liu**, X. Fan, H. Li et al. 2026b, *A JWST/NIRSpec Integral Field Unit Survey of Luminous Quasars at $z \sim 5-6$ (Q-IFU): Rest-frame Optical Nuclear Properties and Extended Nebulae*, ApJ in review, [link](#).
- [9] **W. Liu**, X. Fan, H. Li et al. 2026a, *Frequent Extreme Galaxy-scale Outflows among Luminous Early Quasars*, Nature Accepted, sole 1st and corresp. author, [link](#).
- [8] **W. Liu**, S. Veilleux, S. Sankar et al. 2025, *First Results from the JWST Early Release Science Program Q3D: A Dusty Outflow in a Red Quasar at $z=0.44$* , ApJ, 980, 31, [link](#).
- [7] **W. Liu**, X. Fan, J. Yang et al. 2024b, *Fast Outflow in the Host Galaxy of the Luminous $z=7.5$ Quasar J1007+2115*, ApJ, 976, 33, [link](#).
- [6] **W. Liu**, S. Veilleux, G. Canalizo, et al. 2024a, *Fast Outflows and Luminous He II Emission in Dwarf Galaxies with Active Galactic Nuclei (AGN)*, ApJ, 965, 152, [link](#).
- [5] S. Veilleux, **W. Liu (corresponding author)**, D. Rupke, et al. 2023, *First Results from the JWST Early Release Science Program Q3D: The Warm Ionized Gas Outflow in $z \sim 1.6$ Quasar XID 2028 and Its Impact on the Host Galaxy*, ApJ, 953, 56, [link](#).
- [4] **W. Liu**, S. Veilleux, D. Rupke, et al. 2022, *Galactic Winds across the Gas-Rich Merger Sequence: II. Lyman Alpha Emission and Highly Ionized Outflows in the QUEST ULIRGs*, ApJ, 934, 160, [link](#).
- [3] **W. Liu**, S. Veilleux, G. Canalizo, et al. 2020, *Integral-Field Spectroscopy of Fast Outflows in Dwarf Galaxies with AGN*, ApJ, 905, 166, [link](#).
- [2] **W. Liu**, S. Veilleux, K. Iwasawa, et al. 2019, *Elliptical Galaxy in the Making: The Dual Active Galactic Nuclei and Metal-enriched Halo of Mrk 273*, ApJ, 872, 39, [link](#).
- [1] **W. Liu** and L. Hao 2017, *Study of the Photoionization Properties of Galaxies in the Mid-infrared*, Progress in Astronomy, Vol. 35, No. 2, [link](#).
- Others as Primary Authors:** [7] H. Mao, J. Li, X. Yu, Z. Qu, **W. Liu**, and L. Ji, 2024, *Probing the He II re-ionization ERe via Absorbing C IV Historical Yield (HIERACHY) IV: A complex redshifted absorption system intrinsic to quasar*, ApJ accepted, [link](#).
- [6] S. Sankar, N. L. Zakamska, D. Rupke, **W. Liu**, et al. 2024, *First results from the JWST Early Release Science Program Q3D: AGN photoionization and shock ionization in a red quasar at $z = 0.4$* , submitted, [link](#).
- [5] M. C. Chen, H. Chen, M. Rauch, A. Vayner, **W. Liu**, et al. 2025, *Resolving turbulence drivers in luminous obscured quasars with JWST/NIRSpec IFU*, ApJL, 978, L18, [link](#).
- [4] J. Seebeck, S. Veilleux, **W. Liu**, et al. 2023, *Combined JWST-MUSE Integral Field Spectroscopy of the Most Luminous Quasar in the Local Universe, PDS 456*, ApJ accepted, [link](#).
- [3] A. Aravindan, **W. Liu**, G. Canalizo, et al. 2023, *A Comparison of Outflow Properties in AGN Dwarfs versus Star-Forming Dwarfs*, ApJ, 950, 33, [link](#).
- [2] S. Veilleux, D. Rupke, **W. Liu**, et al. 2021, *Galactic Winds across the Gas-Rich Merger Sequence: I. Highly Ionized N V and O VI Outflows in the QUEST Quasars*, ApJ, 926, 60, [link](#).

- [1] T. Bohn, G. Canalizo, S. Veilleux, and **W. Liu** 2021, *Near-Infrared Coronal Line Observations of Dwarf Galaxies Hosting AGN-driven Outflows*, ApJ, 911, 70, [link](#).
- Others:** [35] X. Lin, X. Fan, Z. Cai, et al. 2025 (**incl. W. Liu**), *The Discovery of Little Red Dots in the Local Universe: Signatures of Cool Gas Envelopes*, submitted, [link](#).
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- [34] T. S. Tanaka, H. B. A., Y. Harikane, et al. 2025 (**incl. W. Liu**), *Discovery of a Little Red Dot candidate at $z \gtrsim 10$ in COSMOS-Web based on MIRI-NIRCam selection*, submitted, [link](#).
- [34] F. D'Eugenio, J. M. Helton, K. Hainline, et al. 2025 (**incl. W. Liu**), *JADES and SAPPHIRES: Galaxy Metamorphosis Amidst a Huge, Luminous Emission-line Region*, MNRAS.tmp.1111D, [link](#).
- [33] M. Pudoka, F. Wang, X. Fan, et al. 2025 (**incl. W. Liu**), *Lyman-break Galaxies in the Megaparsec-scale Environments around Three $z \sim 7.5$ Quasars with JWST Imaging*, ApJ, 987, 198, [link](#).
- [32] S. Fu, F. Sun, L. Jiang, et al. 2025 (**incl. W. Liu**), *Medium-band Astrophysics with the Grism of NIRCam In Frontier Fields (MAGNIF): Spectroscopic Census of $H\alpha$ Luminosity Functions and Cosmic Star Formation at $z \sim 4.5$ and 6.3* , ApJ, 987, 186, [link](#).
- [31] Z. Li, Z. Cai, X. Wang, et al. 2025 (**incl. W. Liu**), *A 13-Billion-Year View of Galaxy Growth: Metallicity Gradient Evolution from the Local Universe to $z = 9$ with JWST and Archival Surveys*, submitted, [link](#).
- [30] Y. Chen, N. L. Zakamska, A. Vayner, et al. 2025 (**incl. W. Liu**), *JWST IFU observations uncover host galaxy continua in extremely red and obscured quasars*, submitted, [link](#).
- [29] F. Sun, J. Yang, F. Wang, et al. 2025 (**incl. W. Liu**), *The Identification of Two JWST/NIRCam-Dark Starburst Galaxies at $z = 6.6$ with ALMA*, submitted, [link](#).
- [28] T. Y. Hsiao, F. Sun, X. Lin, et al. 2025 (**incl. W. Liu**), *SAPPHIRES: Extremely Metal-Poor Galaxy Candidates with $12 + \log(O/H) < 7.0$ at $z \sim 5 - 7$ from Deep JWST/NIRCam Grism Observations*, submitted, [link](#).
- [27] X. Lin, X. Fan, F. Wang, et al. 2025 (**incl. W. Liu**), *Bridging Quasars and Little Red Dots: Insights into Broad-Line AGNs at $z = 5 - 8$ from the First JWST COSMOS-3D Dataset*, submitted, [link](#).
- [26] Y. Zhu, M. J. Rieke, L. C. Ho, et al. 2025 (**incl. W. Liu**), *Nuclear Winds Drive Cold Gas Outflows on Kiloparsec Scales in Reionization-Era Quasars*, submitted, [link](#).
- [25] Y. Fudamoto, J. M. Helton, X. Lin, et al. 2025 (**incl. W. Liu**), *SAPPHIRES: A Galaxy Over-Density in the Heart of Cosmic Reionization at $z = 8.47$* , submitted, [link](#).
- [24] F. Sun, Y. Fudamoto, X. Lin, et al. 2025 (**incl. W. Liu**), *JSlitless Areal Pure-Parallel High-Redshift Emission Survey (SAPPHIRES): Early Data Release of Deep JWST/NIRCam Images and Spectra in MACS J0416 Parallel Field*, submitted, [link](#).
- [23] K. Kakiichi, X. Jin, F. Wang, et al. 2025 (**incl. W. Liu**), *JWST ASPIRE: How Did Galaxies Complete Reionization? Evidence for Excess IGM Transmission around $[O III]$ Emitters during Reionization*, submitted, [link](#).
- [22] J. Lyu, G. H. Rieke, M. Stone, et al. 2025 (**incl. W. Liu**), *Fading Light, Fierce Winds: JWST Snapshot of a Sub-Eddington Quasar at Cosmic Dawn*, ApJL, 981, L20, [link](#).
- [21] Zou, S., Simcoe, R. A., Petitjean, P., et al. 2025 (**incl. W. Liu**), *Disturbed cold gas in galaxy and structure formation*, submitted, [link](#).

- [20] J. Li, X. Yu, H. Mao, et al. 2025 (**incl. W. Liu**), *Probing the He II re-ionization ERA via Absorbing C IV Historical Yield (HIERACHY). II. Project Design, Current Status, and Examples of Initial Data Products*, ApJ, 985, 63, [link](#).
- [19] F. Sun, F. Wang, J. Yang et al. 2025 (**incl. W. Liu**), *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): Spectroscopically Complete Census of Obscured Cosmic Star Formation Rate Density at $z = 4 - 6$* , ApJ, 980, 12, [link](#).
- [18] K. Y. Dan, J. Seebeck, S. Veilleux et al. 2025 (**incl. W. Liu**), *JWST Discovery of a Very Fast Biconical Outflow of Warm Molecular Gas in the Nearby Ultra-Luminous Infrared Galaxy F08572+3915 NW*, ApJ, 979, 68, [link](#).
- [17] J. Champagne, F. Wang, J. Yang et al. 2025 (**incl. W. Liu**), *A Quasar-Anchored Protocluster at $z=6.6$ in the ASPIRE Survey: II. An Environmental Analysis of Galaxy Properties in an Overdense Structure*, ApJ, 981, 114, [link](#).
- [16] J. Champagne, F. Wang, J. Yang et al. 2025 (**incl. W. Liu**), *A Quasar-Anchored Protocluster at $z=6.6$ in the ASPIRE Survey: I. Properties of [OIII] Emitters in a 10 Mpc Overdensity Structure*, ApJ, 981, 113, [link](#).
- [15] X. Jin, X. Fan, J. Yang et al. 2024 (**incl. W. Liu**), *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): JWST Supports Earlier Reionization around [OIII] Emitters*, ApJ accepted, [link](#).
- [14] X. Lin, F. Wang, X. Fan, et al. 2024 (**incl. W. Liu**), *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): Broad-line AGN at $z=4-5$ revealed by JWST/NIRCam WFSS*, ApJ, 974, 147, [link](#).
- [13] R. Decarli, F. Loiacono, E. P. Farina, et al. 2024 (**incl. W. Liu**), *A quasar-galaxy merger at $z \sim 6.2$: Rapid host growth via the accretion of two massive satellite galaxies*, A&A, 689A, 219D, [link](#).
- [12] F. Loiacono, R. Decarli, M. Mignoli, et al. 2024 (**incl. W. Liu**), *A quasar-galaxy merger at $z \sim 6.2$: Black hole mass and quasar properties from the NIRSpect spectrum*, A&A, 685A, 121L, [link](#).
- [11] Y.-C. Chen, Y. Ishikawa, N. L. Zakamska, et al. 2024 (**incl. W. Liu**), *VODKA-JWST: A 3.8 kpc Dual Quasar at Cosmic Noon in a Powerful Starburst Galaxy with JWST/MIRI Integral Field Unit*, ApJ, 968, 92, [link](#).
- [10] C. Bertemes, D. Wylezalek, D. S. N. Rupke et al. 2024 (**incl. W. Liu**), *JWST ERS Program Q3D: The pitfalls of virial BH mass constraints shown in a $z = 3$ quasar with an ultramassive host*, A&A 693A.176B, [link](#).
- [9] D. S. N. Rupke, D. Wylezalek, N. L. Zakamska, et al. 2023 (**incl. W. Liu**), *q3dfit: PSF decomposition and spectral analysis for JWST-IFU spectroscopy*, ascl:2310.004, [link](#).
- [8] Y. Wu, F. Wang, Z. Cai, et al. 2023 (**incl. W. Liu**), *A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): JWST Discovers an Overdensity around a Metal Absorption-selected Galaxy at $z \sim 5.5$* , ApJL, 956, L40, [link](#).
- [7] F. Wang, J. Yang, J. F. Hennawi, et al. 2023 (**incl. W. Liu**), *A SPectroscopic Survey of Biased Halos in the Reionization Era (ASPIRE): JWST Reveals a Filamentary Structure around a $z = 6.61$ Quasar*, ApJL, 951, L4, [link](#).
- [6] J. Yang, F. Wang, X. Fan et al. 2023 (**incl. W. Liu**), *A SPectroscopic Survey of Biased Halos in the Reionization Era (ASPIRE): A First Look at the Rest-Frame Optical Spectra of $z > 6.5$ Quasars Using JWST*, ApJL, 951, L5, [link](#).

- [5] A. Vayner, N. L. Zakamska, Y. Ishikawa et al. 2023 (**incl. W. Liu**), *First Results from the JWST Early Release Science Program Q3D: Powerful Quasar-Driven Galactic Scale Outflow at $z = 3$* , arXiv.2307.13751, [link](#).
- [4] A. Vayner, N. L. Zakamska, Y. Ishikawa, et al. 2023 (**incl. W. Liu**), *First Results from the JWST Early Release Science Program Q3D: Ionization Cone, Clumpy Star Formation, and Shocks in a $z = 3$ Extremely Red Quasar Host*, The ApJ, 955, 92, [link](#).
- [3] D. S. N. Rupke, D. Wylezalek, N. L. Zakamska et al. 2023 (**incl. W. Liu**), *First Results from the JWST Early Release Science Program Q3D: Benchmark Comparison of Optical and Mid-Infrared Tracers of a Dusty, Ionized Red Quasar Wind at $z = 0.435$* , ApJ, vol. 953, Aug. 2023, p. L26., [link](#).
- [2] D. Wylezalek, A. Vayner, D. S. N. Rupke et al. 2022 (**incl. W. Liu**), *First results from the JWST Early Release Science Program Q3D: Turbulent times in the life of a $z \sim 3$ extremely red quasar revealed by NIRSpect IFU*, ApJL, 940, L7, [link](#).
- [1] A. Yazeedi, et al. 2021 (**incl. W. Liu**), *The Impact of Low-luminosity AGNs on Their Host Galaxies: A Radio and Optical Investigation of the Kiloparsec-scale Outflow in MaNGA 1-166919*, ApJ, 916, 102, [link](#).

Successful Telescope Proposals

Selected PI Programs (12 in total, ~ \$276K.)

- JWST **PI**, cycle 2, Survey 3428, 106 hrs, ~ \$120K.
"Quasar Feedback and Supermassive Black Hole Demographics at $z \sim 4-6$ "
- ALMA **PI**, cycle 12, 2025.1.00805.S, 22.7 h hrs.
"The Molecular Phase of the Earliest Galaxy-scale Outflows in Quasars at $z > 6.5$ "
- ALMA **PI**, cycle 10, 2023.1.01135.S, 19.3 hrs.
"The Demographics of Molecular Gas in Dwarf Galaxies with AGN"
- ALMA **PI**, cycle 9, 2022.1.00655.S, Grade C.
"Quasar Feedback in the Reionization Era: A Search for the Molecular Gas Outflow in a $z \sim 6.5$ Broad Absorption Line Quasar"
- HST **PI**, cycle 32, GO 17758, 7 orbits, ~ \$64K.
"Radio Jet-driven AGN Feedback in Dwarf Galaxies: A Pilot Study"
- HST **PI**, cycle 27, GO 15915, 17 orbits, ~ \$81K.
"AGN Feedback in Dwarf Galaxies"
- Keck (NASA) **PI**, 2026A-N119, 1 night, ~ \$11K.
"Probing Extreme Quasar Feedback on the Circumgalactic Medium at $z \sim 5-6$ "
- LBT **PI**, AZ-2023A-014, 2 nights.
"Quasar Feedback in Rapidly Growing, Moderate Mass Black Holes Near Cosmic Noon"
- LBT **PI**, AZ-2024A-011, 2 nights.
"Quasar Feedback in Rapidly Growing, Moderate Mass Black Holes Near Cosmic Noon"
- MMT **PI**, UAO-S183-24B, 2 nights.
"Probing Quasar Feedback on the Circumgalactic Medium at $z \sim 5$ "
- MMT **PI**, UAO-S179-25A, 3.5 nights.
"Probing Quasar Feedback on the Circumgalactic Medium at $z \sim 5$ "
- MMT **PI**, UAO-S157-25B, 3.5 nights.
"LyA Nebulae in $z \sim 5$ Quasars with Galactic-scale Outflows"

Col Programs with Critical Contributions

- JWST **Col**, cycle 1, GO , 2547, 8.7 hrs, PI: S. Veilleux.
 “Quasar Feedback in Action: The Multiphase and Multiscale Outflow of the Most Luminous Quasar in the Local Universe”; **Data Lead.**
- HST **Col**, cycle 33, GO 18134, 59 orbits, PI: S. Veilleux.
 “A Far-Ultraviolet Spectroscopic Survey of AGN Dwarf Galaxies”
- HST **Col**, cycle 29, GO 16698, 25 orbits, PI: S. Veilleux.
 “The Impact of Radio-Mode Feedback on the Circumgalactic Medium of Centaurus A”; **Conceived the project together with S. Veilleux; responsible for technical preparation including target selection, data analysis & visualization.**
- HST **Col**, cycle 26, GO 15662, 50 orbits, PI: S. Veilleux.
 “Galactic Winds across the Gas-Rich Merger Sequence”; **Prepared the proposal together with S. Veilleux; Helped with technical preparation including sample selection.**
- Gemini **Col**, 2019FT, 10.6 hrs, PI: S. Veilleux.
 “AGN Feedback in Dwarf Galaxies”; **Conceived the project and prepared the entire proposal.**

Other Selected Col Programs

- JWST **Col**, cycle 4, GO 7935, 36.3 hrs, PI: F. Sun.
 Efficient Measurement of the Emergence Rate of AGN in Legacy Deep Field
- JWST **Col**, cycle 4, GO 7492, 42.4 hrs, PI: D. Wylezalek.
 MILAN - MIRI LRS Program of AGN at Cosmic Noon
- JWST **Col**, cycle 4, GO 7436, 18.4 hrs, PI: X. Jin.
 The Last Neutral Islands at the End of Reionization? Characterizing the Nature of the Longest Dark Gaps in IGM Transmission at $z \sim 5.3$
- JWST **Col**, cycle 3, GO 5893, 266 hrs, PI: K. Kakiichi.
 COSMOS-3D: A Legacy Spectroscopic/Imaging Survey of the Early Universe
- JWST **Col**, cycle 3, Survey 5645, PI: E. Farina.
 A 3D view of the first QSOs: A JWST/NIRSpec survey program
- JWST **Col**, cycle 3, GO 4691, 16.24 hrs, PI: J. Yang.
 Investigating an Extreme [OIII] Outflow Discovered in a Reionization-era Luminous Quasar
- JWST **Col**, cycle 2, GO 2883, 38.73 hrs, PI: F. Sun.
 MAGNIF: Medium-band Astrophysics with the Grism of NIRCams in Frontier Fields
- JWST **Col**, cycle 2, GO 3325, 44.6 hrs, PI: T. Bohn.
 Big Impact in Little Galaxies? A JWST Investigation of AGN Outflows in Dwarf Galaxies
- JWST **Col**, cycle 2, GO 3663, 21.94 hrs, PI: F. Wang.
 Mapping the Most Extreme Protoclusters in the Epoch of Reionization
- JWST **Col**, cycle 2, GO , 3807, 5.23 hrs, PI: C. Bertemes.
 “Deep grism spectroscopy of the complex environment around an extremely red quasar within an ultramassive host at $z=3$ ”
- JWST **Col**, cycle 2, GO , 3869, 39.23 hrs, PI: S. Veilleux.
 “Extreme Feedback in Action: Fast and Powerful Molecular Outflows in the Local Universe”
- ALMA **Col**, cycle 9, 2022.1.01077.L, PI: F. Wang.
 “A SPectroscopic survey of biased halos In the Reionization Era (ASPIRE): A Quasar Legacy Redshift Survey”
- Keck **Col**, 2020A, 1.5 nights, PI: G. Canalizo.
 “AGN Feedback in Dwarf Galaxies”.
- VLA **Col**, 16B-082 (AG984), 20.5 hrs, PI: J. Gelfand.
 "Radio Emission in Strong Galactic Outflows"

Professional Service & Collaboration:

Referee, *ApJ*, *A&A*, *Nature*.

Chair, *Galaxy Working Group*, *JWST Treasury Program COSMOS-3D*.

Key member, *JWST ERS program "Q3D" collaboration and q3dfit software development team*.

Member, *EREBUS JWST High-z Quasar Collaboration*.

Mentoring

Oct 2024 – **Primary mentor for a sophomore, Nayera Abdessalam, at the University of Arizona**,
Now *Project: "Black Hole Mass of High-z Quasars"*, awarded the W.A. Franke Honors College Exploratory Mini-Grant and a private grant at UA.

May 2021 – **Primary mentor for a high school student, Orion Foo, from Poolesville High School**,
May 2022 *Project: "A Census of P V-traced, AGN-driven Outflows with HST/COS"*, based on this project, Orion was named a Top 300 Scholar in the 81st Regeneron Science Talent Search (STS), the nation's oldest and most prestigious science and mathematics competition for high school seniors. He was also chosen to attend the International Science and Engineering Fair as a regional representative based on the same research.

Teaching & Outreach Experience

Jun. 2024 **Volunteer**, *Black Hole Fusion Camp*, Delivered demonstrations and lessons to 3rd-5th graders at Flandrau Planetarium's Fusion Camp.

Sep. 2018 **Volunteer**, *Maryland Day 2018*, Presented Astronomy Research Demos.

Mar. 2018 **Master of Ceremony**, *UMD Observatory Open House*, Hosted the ceremony.

Spring 2018 **Teaching Assistant**, *"Introductory Astrophysics II"*, Designed and led the discussion sessions and quizzes/exams independently.

Fall 2016 **Teaching Assistant**, *"Stars and Stellar Systems"*.

Fall 2016 **Teaching Assistant**, *"Origin of the Universe"*.

Recent Professional Talks

Jun. 2025 **EREBUS Workshop**, *Italy*, A JWST Spectroscopic Survey of $z \sim 5-6$ Quasars: Extreme Galaxy-scale Outflows and Nuclear Properties.

Oct. 2024 **Tinsley Workshop**, *New Haven*, Probing Quasar-driven Outflows at $z > 6$ with JWST.

Sep. 2024 **The First Gigayear(s)**, *Hilo*, Probing Quasar-driven Outflows at $z > 6$ with JWST.

Jul. 2024 **The Origin and Evolution of Supermassive Black Holes**, *Italy*, Galaxy-scale Outflow in a $z = 7.5$ Quasar Revealed by JWST.

Jul. 2024 **European Astronomical Society Annual Meeting 2024**, *Italy*, Probing Quasar-driven Outflows at $z > 6$ with JWST.

May. 2024 **Extreme Galaxies in Their Extreme Environments at Extremely Early Epochs**, *Iceland*, Galaxy-scale Outflow in a $z = 7.5$ Quasar Revealed by JWST.

Jun. 2023 **Roman Science Inspired by Emerging JWST Results**, *Baltimore*, Unveiling Quasar Host Galaxies at Cosmic Dawn with JWST/IFU.

Oct. 2022 **FLASH Talk**, *NOIRLab*, AGN Feedback in Dwarf Galaxies.

- Sep. 2022 **Origin, Growth and Feedback of Black Holes in Dwarf Galaxies**, *Virtual*, AGN Feedback in Dwarf Galaxies.
- Aug. 2022 **8th Annual GMTO Community Science Meeting: Black Holes at All Scales**, *Sedona*, AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Dec. 2021 **Seminar at UCSC**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Dec. 2021 **Astrocoffee at Institute of Astronomy, U. of Hawaii**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Nov. 2021 **Galaxy Journal Club at STScI**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Nov. 2021 **Monday seminar at UT Austin**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Nov. 2021 **Galread at Princeton**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies.
- Nov. 2021 **Tuesday Lunch Talk at UCLA**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Nov. 2021 **Galaxy Group Seminar at U. of Arizona**, *Virtual*, Probing AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Sep. 2021 **Weekly Science Talk at Caltech/IPAC**, *Virtual*, Exploring AGN Feedback in Dwarf Galaxies and Ultraluminous Infrared Galaxies.
- Aug. 2021 **AGN Discussion Group Seminar at NASA/Goddard**, *Virtual*, AGN Feedback in Dwarfs and ULIRGs.
- May. 2021 **Multi-Object Spectroscopy for Statistical Measures of Galaxy Evolution at STScI**, *Virtual*, Fast, Kpc-scale Outflows in Dwarf Galaxies with AGN.

Data Expertise

Optical Integral Field Spectroscopy: Keck/KCWI, Gemini/GMOS, VLT/MUSE, SDSS IV/MaNGA, VENGA (VIRUS-P at Harlan J. Smith telescope at McDonald Obs).

Infrared Integral Field Spectroscopy: JWST/NIRSPEC IFU, JWST/MIRI MRS.

1-D Optical/Near-infrared Spectroscopy: SDSS, VLT/X-SHOOTER.

Far-ultraviolet Spectroscopy and Imaging: HST/COS, HST/STIS, HST/WFC3.

X-ray Imaging Spectroscopy: Chandra/ACIS.

Slitless Spectroscopy: JWST/NIRCam WFSS.